A Multi GPU fft in MATLAB for the Phase-Field Crystal Framework

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Summary

We present a MATLAB-based framework for multi-GPU fast Fourier transforms (ffts) designed for large-scale pseudo-spectral simulations. The software implements two complementary multi-GPU strategies that overcome single-GPU memory limitations and accelerate spectral solvers. Applied to phase-field crystal (PFC) models and their multiphysics extensions, the framework achieves speedups of approximately six for standard PFC simulations and up to sixty for multiphysics formulations compared to a purely CPU-based implementation.

Statement of need

Large-scale pseudo-spectral simulations are often limited by the memory capacity and performance of single GPUs due to their reliance on repeated multi-dimensional ffts. To address this bottleneck, we present two complementary multi-GPU fft strategies.

The first strategy distributes a single high-dimensional fft across multiple GPUs using domain decomposition and inter-GPU communication, enabling Fourier transforms on problem sizes exceeding individual device memory. The second strategy computes multiple ffts concurrently by assigning different physical fields to separate GPUs with synchronized communication.

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These approaches are essential building blocks for pseudo-spectral solvers in large-scale and multiphysics applications. As a representative use case, we integrate both strategies into a Fourier pseudo-spectral solver for the phase-field crystal (PFC) model, where large spatial domains and coupled fields make single-GPU execution impractical. The algorithms are implemented in MATLAB, providing an accessible and extensible foundation for GPU-accelerated spectral simulations and introducing the first multi-GPU implementation in MATLAB.

Single fft on Multiple GPUs

Distributed multi-GPU ffts based on domain decomposition and inter-GPU communication are well established in high-performance computing; see, e.g., [1, 2, 3, 4]. These approaches enable scalable multi-dimensional ffts on large GPU systems.

In contrast, fft-based pseudo-spectral solvers for the phase-field crystal (PFC) equation have so far been limited to single-GPU implementations [5] or CPU-based parallelization [6]. To our knowledge, a dedicated multi-GPU fft implementation tailored to large-scale three-dimensional PFC simulations has not been reported.

Following the strategy of [1, 2], we decompose a three-dimensional fft of size $N_x \times N_y \times N_z$ across G GPUs by slab decomposition along the z-direction, cf. Fig. 1(a). Each GPU first performs local two-dimensional ffts, followed by peer-to-peer (P2P)communication to redistribute the data, and a final one-dimensional fft along the remaining dimension. After completion, each GPU holds a portion of the transformed Fourier-space array.

We apply this strategy to the PFC equation, which describes the evolution of a periodic density field ψ [7]. For an FCC crystal symmetry, the free-energy functional reads

$$F[\psi] = \int_\Omega \left[\frac{\psi}{2} \left(\varepsilon + (1+\nabla^2)^2 (4/3 + \nabla^2)^2 \right) \psi + \frac{\psi^4}{4} \right] \mathrm{d}\mathbf{r}, \tag{1}$$

with corresponding evolution equation

$$\partial_t \psi = (\varepsilon + \mathcal{L}) \nabla^2 \psi + \nabla^2 \psi^3. \tag{2}$$

The equation is solved using a Fourier pseudo-spectral method with semi-implicit time integration [8], requiring one forward and one backward fft per time step.

We benchmark the multi-GPU solver for problem sizes ranging from 750^3 to 1400^3 using three systems from HPC clusters provided by NHR Center of TU Dresden: GPU computations are performed with four NVIDIA H100 SXM5 GPUs (94 GiB HBM2e each, HPC cluster Capella) and with eight NVIDIA A100 SXM4 GPUs (40 GiB HBM2 each, HPC cluster Alpha Centauri). CPU reference runs are performed on an Intel Xeon Platinum 8470 (100 cores, 2.00 GHz, HPC cluster Barnard).

Multiple GPU usage for Multiphysics PFC

The PFC framework readily supports multiphysics extensions. As an example, we consider the hydrodynamic phase-field crystal (hydrodynamic PFC) model [9] in three dimensions, which augments the density field ψ with a three component mesoscopic velocity field $\mathbf{v} = (v_1, v_2, v_3)$:

$$\begin{aligned} \partial_t \psi &= \nabla^2 \left(\frac{\delta F}{\delta \psi} \right) - \mathbf{v} \nabla \psi, \\ \rho \partial_t \mathbf{v} &= \Gamma \nabla^2 \mathbf{v} - \left\langle \psi \nabla \frac{\delta \mathcal{F}_\psi[\psi]}{\delta \psi} \right\rangle. \end{aligned} \tag{3}$$

We distribute the fields across four GPUs, assigning one field per device and performing synchronized inter-GPU communication after each time step. This strategy enables problem sizes beyond single-GPU memory limits.

Benchmark results using four NVIDIA H100 GPUs demonstrate substantial runtime reductions compared to CPU execution, particularly for large domains where single-GPU simulations are infeasible, see Fig. 1(b).

Availability

Repository: `https://github.com/name/project`
License: Creative Commons Attribution 4.0 International
Zenodo DOI: `https://doi.org/...`

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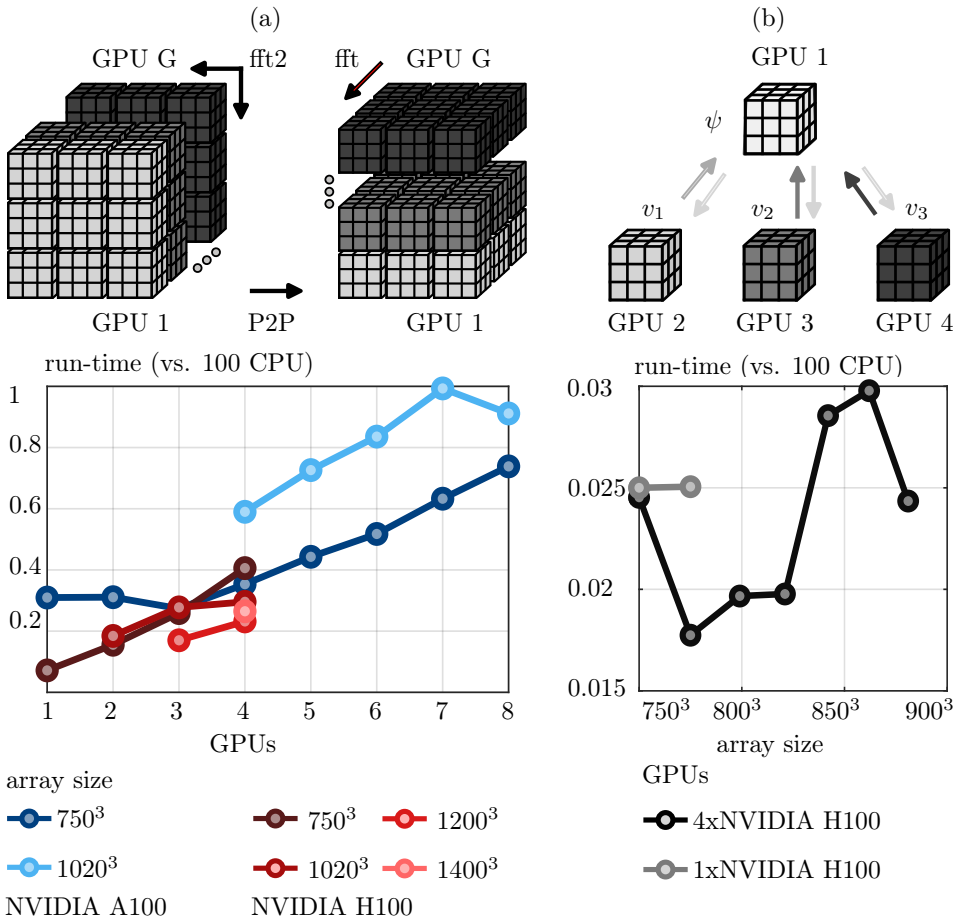

Figure 1: (a) Schematic of the multi-GPU fft algorithm based on slab decomposition for a three-dimensional array. The data are decomposed along the z-direction, followed by local two-dimensional ffts, peer-to-peer communication, and a final one-dimensional fft (upper panel). Relative runtimes for 1000 time steps are shown, normalized by CPU execution time. Speedups of up to a factor of six are observed, with optimal performance on a single GPU for 750^3 and multi-GPU execution required for larger domains due to memory constraints. An array of size 1400^3 fits only on four H100 GPUs and reaches approximately 17 % of the CPU runtime (lower panel). (b) Decomposition of a pseudo-spectral multiphysics PFC solver (e.g., hydrodynamic PFC) across four GPUs (upper panel). Relative runtimes of the hydrodynamic PFC solver are shown as a function of problem size, demonstrating that multi-GPU execution enables simulations up to a problem size of 900^3, which is infeasible on a single GPU. Compared to a CPU implementation, speedups of up to a factor of sixty are achieved (lower panel).